

Erdős Problem #955

EP-955

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $s(n) = \sigma(n) - n = \sum_{\substack{d|n \\ d < n}} d$ be the sum of proper divisors function.

If $A \subset \mathbb{N}$ has density 0 then $s^{-1}(A)$ must also have density 0.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>